

BANDIRMA, Turkey

WMO: 171150

Lat: 40.3162N

Lon: 27.9723E

Elev: 138

StdP: 14.62

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.9	28.4	19.6	14.8	29.7	22.3	16.9	31.7	40.7	34.3	34.1	40.1	6.4	0	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.5	89.9	69.1	87.5	69.2	85.2	69.4	74.7	82.4	73.5	81.2	72.4	80.5	11.2	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.9	118.7	79.3	71.0	114.8	78.8	69.6	109.4	78.3	38.3	82.5	37.1	81.3	36.2	81.0	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.6	25.7	23.0	20.2	98.3	4.5	5.1	16.9	102.0	14.3	104.9	11.7	107.8	8.4	111.5		
			DB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB		
			19.4	76.5	4.5	1.7	16.1	77.7	13.5	78.7	10.9	79.7	7.6	80.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.5	41.8	43.6	47.5	54.0	62.6	71.2	76.3	77.0	69.3	60.8	52.1	45.0
	DBStd	13.64	6.90	7.10	6.63	6.08	5.16	4.54	3.37	3.57	5.10	5.96	6.81	6.96
	HDD50	870	266	201	131	26	0	0	0	0	3	54	189	
	HDD65	3493	718	599	544	333	108	6	0	19	155	390	621	
	CDD50	3983	13	22	52	147	391	637	816	837	580	338	117	33
	CDD65	1130	0	0	1	3	33	193	351	372	149	25	3	0
	CDH74	8986	0	0	6	58	357	1535	2979	3011	899	128	13	0
	CDH80	2464	0	0	0	9	73	443	900	809	205	25	0	0
Wind	WSAvg	10.3	10.1	11.0	10.2	8.7	8.4	9.6	11.9	13.2	10.8	10.4	8.6	10.2
Precipitation	PrecAvg	28.5	4.4	3.0	2.8	2.4	1.5	1.1	0.6	0.7	1.3	2.6	3.7	5.0
	PrecMax	42.8	13.1	8.3	6.7	8.3	4.8	4.3	3.3	4.6	6.1	8.1	14.3	13.7
	PrecMin	19.0	0.5	0.2	0.3	0.3	0.2	0.0	0.0	0.0	0.0	0.9	0.4	
	PrecStd	5.6	3.0	1.8	1.7	1.5	1.1	0.8	1.1	1.5	1.9	2.4	2.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.0	67.6	73.2	80.6	86.4	93.5	95.2	93.2	90.0	82.8	75.5	67.0
		MCWB	53.9	55.4	57.7	62.7	65.6	68.0	69.2	69.7	67.2	64.3	61.5	56.9
	2%	DB	59.4	62.6	67.9	74.3	81.3	87.9	89.9	89.1	85.6	77.2	70.0	62.7
		MCWB	52.0	53.2	55.3	59.8	64.8	67.2	68.8	71.1	66.8	64.3	59.3	54.5
	5%	DB	55.8	59.0	64.1	70.1	78.2	84.6	86.8	86.2	81.2	73.3	66.2	59.2
		MCWB	49.9	51.4	53.6	58.0	63.3	67.4	69.5	71.5	67.1	63.2	57.8	52.9
	10%	DB	52.6	55.2	59.3	66.2	74.4	82.0	84.5	84.3	78.9	70.0	62.7	55.5
		MCWB	47.8	49.3	51.5	56.2	62.0	67.3	69.4	71.2	67.0	62.2	56.5	50.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.2	56.5	59.0	63.8	68.6	73.7	75.3	76.7	73.1	68.4	63.6	57.9
		MCDB	61.8	64.8	71.2	77.9	80.8	81.7	83.7	84.4	79.4	75.2	71.7	65.2
	2%	WB	52.5	53.9	56.0	60.8	66.4	71.7	74.0	75.3	71.0	66.1	60.4	55.2
		MCDB	58.5	61.4	65.5	72.0	77.8	80.7	82.4	82.9	78.2	73.6	67.4	61.2
	5%	WB	50.4	51.6	54.1	58.7	64.6	70.1	72.9	74.2	69.7	64.5	58.6	53.2
		MCDB	55.5	58.0	62.7	68.9	75.0	79.3	81.3	81.7	77.3	71.7	64.5	58.6
	10%	WB	48.3	49.6	51.9	56.6	63.1	68.9	71.9	73.1	68.5	63.0	56.7	51.2
		MCDB	52.3	54.8	59.1	65.3	72.2	78.6	80.6	80.6	76.3	69.2	62.4	55.4
Mean Daily Temperature Range	5% DB	MDBR	12.8	13.8	16.4	18.1	18.9	18.5	16.3	14.5	16.7	15.0	15.7	12.7
		MCDBR	19.5	22.1	26.0	28.0	28.1	25.8	21.9	18.4	23.3	22.3	22.6	17.6
		MCWBR	13.6	14.4	15.6	16.2	14.4	11.3	9.5	7.7	11.3	12.1	14.2	12.0
	5% WB	MCDBR	17.9	19.9	23.6	25.0	23.4	19.7	15.4	14.1	17.2	17.4	19.1	16.1
		MCWBR	12.8	13.3	14.8	15.1	12.8	9.4	7.5	6.5	9.4	10.2	12.6	11.4
Clear-Sky Solar Irradiance	taub	0.369	0.400	0.437	0.479	0.445	0.429	0.432	0.440	0.429	0.415	0.383	0.372	
	taud	2.323	2.216	2.120	2.028	2.203	2.292	2.293	2.268	2.283	2.289	2.324	2.316	
	Ebn at Noon	244	253	256	253	266	270	268	262	257	245	237	232	
	Edh at Noon	29	37	45	53	45	42	41	41	38	34	29	27	
All-Sky Solar Radiation	RadAvg	508	729	1144	1590	2020	2286	2385	2064	1542	1006	652	468	
	RadStd	71	99	110	178	151	159	79	93	108	93	83	68	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.76	N/A	+1.84	N/A	+1.15	+1.46	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	+1.01	N/A	N/A	+1.04	N/A	N/A	N/A	-227	+247	+122

Nomenclature: See separate page